

Curvature Polarization Transport Gravity: Preliminary Figure-Digitized Stress Test of AGC 114905

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Abstract

AGC 114905 is a gas-rich ultra-diffuse galaxy with unusually low circular speed for its baryonic mass. It is therefore a useful stress test for Curvature Polarization Transport Gravity (CPTG), because the target behavior is not a large halo-like enhancement but a controlled low-response branch. This note presents a preliminary CPTG comparison using figure-digitized radial data and digitized velocity uncertainties from the 2024 AGC 114905 analysis. The calculation compares a baryon-only curve, a standard-acceleration MOND curve, and a reduced CPTG galaxy-response curve on the same five digitized radial points, with residuals reported relative to the supplied observational uncertainty estimates. Within this five-point diagnostic, the reduced CPTG response tracks the digitized observed curve with an RMS residual of 1.7427 km s^{-1} , while the baryon-only curve underpredicts and the standard MOND curve overpredicts the same target. A 1000-draw robustness audit preserves the CPTG preference under local digitization, baryonic-curve, and velocity perturbations. The appropriate claim level is a preliminary figure-digitized stress-test pass, not final radial validation, until the digitized rows are replaced by author-supplied or externally digitized radial tables.

1 Motivation

AGC 114905 is a gas-rich ultra-diffuse galaxy whose rotation curve has been used as a stress test for dark-matter and modified-gravity interpretations. The 2022 resolved H I analysis reported a regular disc rotating at about 23 km s^{-1} , with the flat part of the rotation curve reached and the baryonic mass distribution explaining the circular speed almost entirely within the observed radius [1]. The 2024 ultra-deep optical and H I reanalysis found an inclination of $31 \pm 2^\circ$, reported a flat rotation speed near 25 km s^{-1} , and concluded that baryons alone do not provide enough gravitational contribution, while standard MOND also does not fit the object [2].

For CPTG [5, 6], the important feature is that AGC 114905 is not a conventional missing-mass target. The galaxy tests whether a baryon-sourced curvature response can remain low in a diffuse gas-rich system while still providing the mild

residual support required by the updated rotation curve. The object therefore probes a suppressed or low-response galaxy branch rather than an ordinary high-enhancement dwarf branch.

2 Input status

The calculation in this note uses a five-point table digitized from the 2024 mass-model figure, where the authors compare the observed circular-speed profile with the no-dark-matter baryonic contribution and related model curves [2]. The digitization was performed from a 300 dpi PDF render of the plotted panel. The velocity uncertainties used in the residual and chi-square calculations are digitized estimates from the plotted observational error bars in the same published comparison figure [2]. They should be treated as figure-level observational uncertainty estimates, not as a substitute for the full covariance or author-supplied machine-readable radial table. The input rows are therefore suitable for a preliminary response test, but they are not a substitute for author-supplied machine-readable radial tables.

The working radial quantities follow the rotation-curve and baryonic-decomposition quantities used in the AGC 114905 mass-model analyses [1, 2]:

$$R, \quad V_{\text{circ}}(R), \quad \sigma_V(R), \quad V_{\star}(R), \quad V_{\text{gas}}(R), \quad V_{\text{bar}}(R). \quad (1)$$

The ordinary baryonic speed in the comparison panel is treated with the same quadrature convention used for the plotted stellar and gas contributions in the AGC 114905 mass-model decomposition [1, 2]:

$$V_{\text{bar}}^2(R) = V_{\text{gas}}^2(R) + V_{\star}^2(R). \quad (2)$$

R kpc	Observed $V_{\text{circ}} \pm \sigma_V$	$V_{\star}$	V_{gas}	V_{bar}
1.1264	9.9363 ± 1.9108	6.1704	2.9459	6.8471
3.3767	16.8232 ± 1.9108	8.9968	8.8376	12.5796
5.6258	22.7309 ± 1.8710	8.2006	13.9331	16.1624
7.8761	24.6736 ± 2.2691	7.6831	17.5955	19.1879
10.1252	24.2516 ± 1.9506	6.8471	19.5860	20.7803

Table 1: Digitized radial inputs used in the preliminary AGC 114905 stress test. All velocities are in km s^{-1} . The input quantities are read from the 2024 AGC 114905 mass-model comparison figure [2]. The σ_V values are observational error-bar estimates read from the plotted 2024 mass-model panel and are used in the residual and χ^2 calculations.

The observed response target is

$$\mathcal{R}_{\text{obs}/\text{bar}}(R) = \frac{g_{\text{obs}}(R)}{g_{\text{bar}}(R)} = \frac{V_{\text{circ}}^2(R)}{V_{\text{bar}}^2(R)}. \quad (3)$$

3 Claim level

This note is a preliminary figure-digitized CPTG stress test. The result tests whether the reduced CPTG response equation can reproduce the plotted AGC 114905 velocity scale better than the same-table baryon-only and standard-MOND comparisons. It does not claim final radial validation, does not replace the published tilted-ring modeling used to infer the rotation curve [1, 2], and does not re-fit CDM, SIDM, or FDM halo models. Those comparisons remain literature-context comparisons from the published AGC 114905 mass-model analysis until the corresponding radial model chains or machine-readable profiles are reconstructed [2].

The useful current claim is therefore limited and specific: on the five digitized points used here, CPTG provides a low-response branch that tracks the observed circular-speed curve while avoiding the large MOND-like overprediction reported as a central difficulty for AGC 114905 under the measured geometry [1, 2]. This is the correct target behavior for a diffuse, gas-rich UDG whose rotation curve lies close to the baryonic contribution but is not exactly baryon-only under the updated 2024 geometry.

4 Reduced CPTG galaxy-response calculation

The reduced CPTG source speed used in this preliminary calculation is the CPTG-weighted gas-plus-stellar source. This is a CPTG modeling choice, not part of the published AGC 114905 observational analysis; the observational inputs are the gas, stellar, and circular-speed profiles from the published mass-model figures [2]:

$$V_{\text{bar},C}^2(R) = \frac{27}{50}V_{\text{gas}}^2(R) + \frac{23}{50}V_{\star}^2(R). \quad (4)$$

This gives the CPTG source acceleration

$$g_{\text{bar},C}(R) = \frac{V_{\text{bar},C}^2(R)}{R}. \quad (5)$$

The solved reduced CPTG acceleration is written as

$$g_C(R) = g_{\text{bar},C}(R) + \frac{\sqrt{a_{\star}g_{\text{bar},C}(R)} + g_{\text{geom}}(R)}{\left[1 + (g_C(R)/a_{\star})^{123/50}\right]^{11/50}}. \quad (6)$$

The geometric term is

$$g_{\text{geom}}(R) = \eta a_{\star} \frac{(R/R_c)^{7/5}}{1 + (R/R_c)^{12/5}}, \quad (7)$$

with

$$\eta = \frac{1}{2}, \quad R_c = \frac{48\pi c^2}{a_{\star}}. \quad (8)$$

The fitted velocity is

$$V_{\text{CPTG}}(R) = \sqrt{g_C(R)R}. \quad (9)$$

For the five digitized points, the best-fit response scale is

$$a_\star = 3.981381414489 \times 10^{-12} \text{ m s}^{-2}. \quad (10)$$

5 Comparison models

The baryon-only comparison uses the digitized red baryonic curve directly from the published AGC 114905 mass-model figure [2]. The MOND comparison uses the same digitized baryonic curve and the standard acceleration scale commonly used in MOND phenomenology [3, 4]:

$$a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}. \quad (11)$$

The MOND curve is computed as a direct algebraic comparison curve using the same baryonic input, following the usual MOND-style relation between baryonic and total acceleration [3, 4]:

$$V_{\text{MOND}}^2(R) = V_{\text{bar}}^2(R) + \frac{V_{\text{bar}}^2(R)}{2} \left[\sqrt{1 + \frac{4a_0 R}{V_{\text{bar}}^2(R)}} - 1 \right]. \quad (12)$$

The comparison statistic uses the digitized observational velocity uncertainties from Table 1:

$$\chi^2 = \sum_i \left[\frac{V_{\text{model}}(R_i) - V_{\text{circ}}(R_i)}{\sigma_{V,i}} \right]^2. \quad (13)$$

A diagnostic BIC-like value is also reported as

$$\text{BIC}_{\text{diag}} = \chi^2 + k \ln n, \quad (14)$$

where $n = 5$ and $k = 1$ for the reduced CPTG response fit. This statistic is only a compact diagnostic on the digitized five-row table; it is not the full BIC analysis of the observational paper. The CDM, SIDM, and FDM statements in this note are not new fits; they are carried only as context from the published AGC 114905 analysis [2].

6 Observed versus CPTG

R kpc	Observed $V_{\text{circ}} \pm \sigma_V$	CPTG V	CPTG–Obs.	Residual	CPTG/Obs.
1.1264	9.9363 ± 1.9108	8.6731	-1.2632	-0.6611σ	0.8729
3.3767	16.8232 ± 1.9108	15.8431	-0.9802	-0.5129σ	0.9417
5.6258	22.7309 ± 1.8710	20.6109	-2.1200	-1.1331σ	0.9067
7.8761	24.6736 ± 2.2691	24.5523	-0.1213	-0.0534σ	0.9951
10.1252	24.2516 ± 1.9506	27.1010	2.8494	1.4608σ	1.1175

Table 2: Observed versus reduced CPTG velocity comparison for the five digitized AGC 114905 radial points. Velocities are in km s^{-1} . The observed uncertainty column is digitized from the plotted 2024 AGC 114905 mass-model comparison [2] and is included explicitly because it is the denominator used for the residual significances and the χ^2 statistic. This table should be replaced with author-table values when available.

The reduced CPTG fit has

$$\text{RMS} = 1.7427 \text{ km s}^{-1}, \quad \langle |\Delta V| \rangle = 1.4668 \text{ km s}^{-1}, \quad \left\langle \frac{V_{\text{CPTG}}}{V_{\text{obs}}} \right\rangle = 0.9668. \quad (15)$$

The inner and mid-disc points are mildly low, the point near 7.9 kpc is nearly exact, and the outer point is high by 2.8494 km s^{-1} .

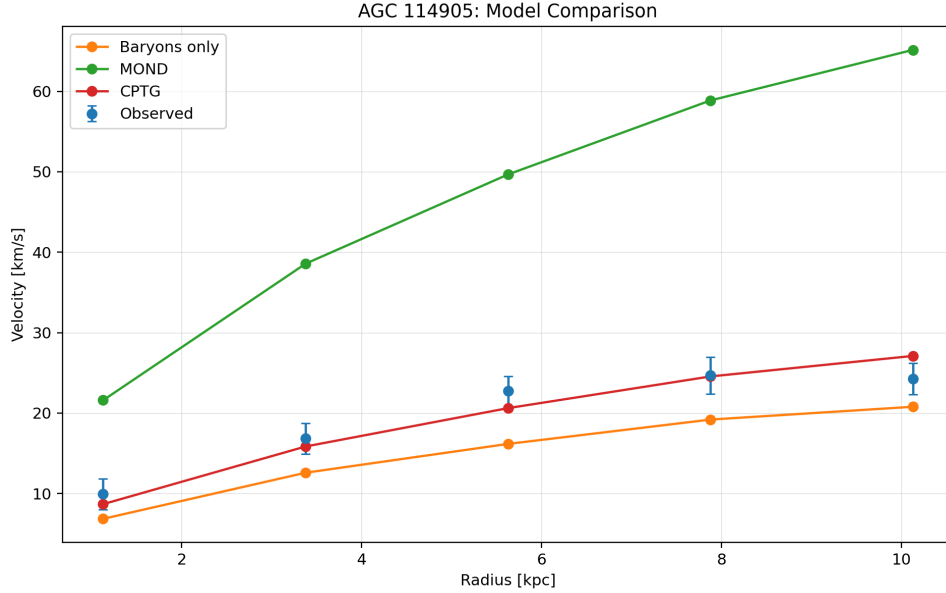


Figure 1: Observed curve, digitized baryon-only curve, standard MOND curve, and reduced CPTG response curve on the same five digitized points. The observational and baryonic curves are digitized from the published 2024 AGC 114905 mass-model comparison [2]. The plot is a preliminary response comparison, not a replacement for full mass-model chains.

7 Model comparison

Model	χ^2	RMS residual	Mean $ \Delta V $	Max residual
Baryon-only digitized curve	28.8818	4.7506	4.5717	3.5106σ
MOND, standard a_0	1040.9708	28.9048	27.0900	20.9698σ
Reduced CPTG response	4.1206	1.7427	1.4668	1.4608σ

Table 3: Digitized five-point model comparison. Velocity residuals are in km s^{-1} . The comparison uses one fitted reduced-CPTG response scale; baryon-only and MOND are direct same-table diagnostics based on the published AGC 114905 baryonic decomposition [2].

On the five-point digitized table, the baryon-only curve underpredicts the observed speeds, the standard MOND curve strongly overpredicts the observed speeds, and the reduced CPTG response tracks the curve with the lowest residuals. This behavior is consistent with the 2024 paper’s statement that baryons alone are insufficient and standard MOND does not fit AGC 114905 under the measured geometry [2].

8 Robustness audit

The robustness audit perturbed the digitized velocities, baryonic curve, radii, gas contribution, and stellar contribution from the figure-derived input table, then re-fit the reduced CPTG response scale. The audit used 1000 Monte Carlo draws. The result was

$$N_{\text{draws}} = 1000, \quad N_{\text{CPTG wins}} = 1000. \quad (16)$$

Quantity	p16	p50	p84
$\Delta\text{BIC}_{\text{CPTG}-\text{baryons}}$	-28.7455	-22.6809	-16.7075
$\Delta\text{BIC}_{\text{CPTG}-\text{MOND}}$	-1040.3747	-984.3531	-929.5508
$a_\star$	3.5632×10^{-12}	3.9396×10^{-12}	4.4206×10^{-12}
CPTG RMS residual	1.5580	1.8913	2.2670

Table 4: Monte Carlo robustness summary. Negative diagnostic BIC differences favor CPTG after the one-parameter penalty. RMS residuals are in km s^{-1} .

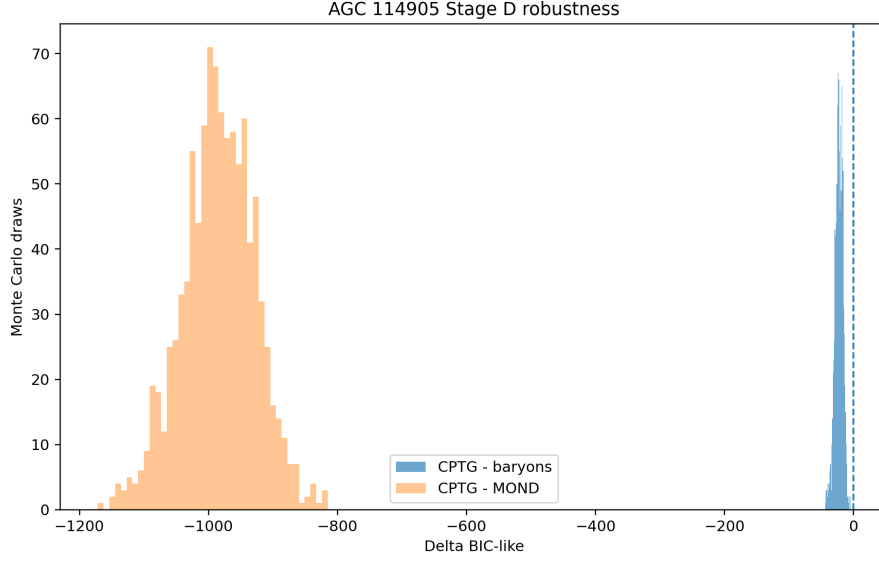


Figure 2: Distribution of diagnostic BIC-like differences across the 1000-draw robustness audit. Negative values favor the reduced CPTG response after the one-parameter penalty. The distribution quantifies local digitization robustness only.

The inclination branch audit rescales the observed velocity branch while keeping the digitized baryonic curve fixed. It is therefore a diagnostic rescaling, not a new tilted-ring or surface-density reconstruction. The inclination sensitivity is important because both AGC 114905 papers identify inclination as the central geometric uncertainty in the mass-model interpretation [1, 2]. Across the tested branches, the reduced CPTG response remains the BIC-like winner. The physical branch favored by the 2024 optical analysis remains near 31° ; the very low-inclination branch needed by standard MOND is reported as inconsistent with the observed morphology [2].

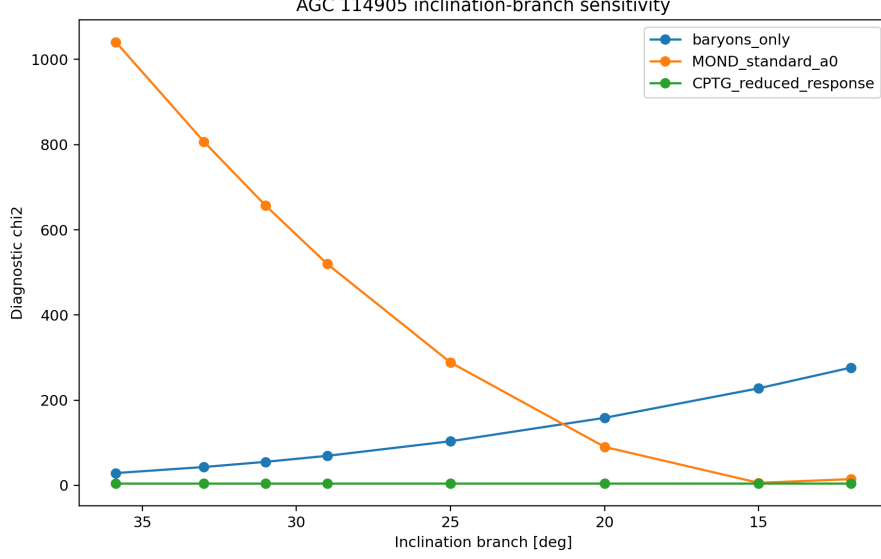


Figure 3: Inclination-branch diagnostic. Each branch rescales the observed velocity curve while preserving the same digitized baryonic model. This is a sensitivity check, not a new H I cube or baryonic-profile reconstruction at each inclination.

9 Interpretation

The preliminary CPTG result identifies AGC 114905 as a low- a_* suppressed-response branch. This interpretation is motivated by the published finding that AGC 114905 sits close to the baryonic contribution but remains difficult for standard MOND and ordinary CDM halo expectations under the measured geometry [1, 2]. The fitted acceleration scale is approximately

$$a_* \approx 4 \times 10^{-12} \text{ m s}^{-2}, \quad (17)$$

which is far below ordinary galaxy-scale acceleration anchors used in broader CPTG work. This is the expected qualitative direction for a diffuse gas-rich UDG whose observed rotation curve lies close to the baryonic curve but still requires a mild lift under the updated 2024 mass model.

The result is not a halo fit. The calculation uses the baryonic radial structure as the source and solves a reduced CPTG response branch. The comparison shows that, on the present five-point digitized table, the response can stay low enough to avoid a MOND-like overprediction while still lifting the baryonic curve toward the observed circular speeds.

10 Data status and next steps

The calculation should be upgraded in the following order:

1. Replace the figure-digitized rows and digitized error-bar estimates with author-supplied or externally digitized radial tables, including the observational uncertainty model from the AGC 114905 analyses [1, 2].
2. Rebuild the baryonic source directly from the gas and stellar surface-density profiles.
3. Refit the reduced CPTG response using the full radial table rather than five plotted points.
4. Add a formal inclination and distance branch grid tied to reconstructed baryonic profiles.
5. Compare against the published CDM, SIDM, and FDM mass-model chains only after their radial models are reconstructed or supplied.

Until those steps are completed, the appropriate claim level is a preliminary figure-digitized CPTG stress-test pass for AGC 114905, not a final radial validation.

11 Conclusion

AGC 114905 is a useful CPTG stress test because it requires a weak, controlled curvature response rather than a large dark-halo-like enhancement. Using the improved five-point digitization of the 2024 baryonic mass-model figure, including digitized observational velocity error bars, the reduced CPTG galaxy-response equation fits the observed circular speeds with an RMS residual of 1.7427 km s^{-1} . The same digitized table gives a baryon-only underprediction and a strong standard-MOND overprediction. A 1000-draw robustness audit preserves the CPTG preference under local perturbations of the digitized data. Within the limits of the figure-digitized table, the result supports AGC 114905 as a candidate low-response CPTG structural branch and motivates a full radial analysis using author-supplied or externally digitized data.

References

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